Matrix functions and stability

Decidability of strict stability for rational matrix families

Difficulty: extreme **Importance:** broadly interesting

Status: open; book question with subsequent-literature screening

Context and notation

The joint spectral radius of a nonempty compact set $\mathcal{M} \subset \mathbb{C}^{d\times d}$ is

$$\hat{\rho}(\mathcal{M}) = \lim_{k\to\infty} \max_{A_1,\dots,A_k\in\mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

This definition also applies to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

Does a Turing machine exist which, on every input consisting of a finite nonempty list of matrices in $\mathbb{Q}^{d\times d}$, encoded by binary integer numerators and positive denominators, halts and correctly decides whether

$$\hat{\rho}(\mathcal{M}) < 1?$$

Dimension and list length are part of the input. No separation from the threshold is promised. The question asks for termination and correctness, without a polynomial running-time requirement.

References and status evidence

Jungers, *The Joint Spectral Radius: Theory and Applications*, §2.2.3, Open Question 1, printed p. 29 of the author manuscript. Blondel and Tsitsiklis, The boundedness of all products of a pair of matrices is undecidable, Systems & Control Letters 41 (2000), 135–140, §2, proves undecidability of the non-strict test $\hat{\rho} \le 1$. That theorem does not answer the strict test.

The screen included "joint spectral radius" with "strict", "decidability", "less than one", and 2025/2026. No deciding algorithm or applicable undecidability reduction was located. Some later literature summarizes the non-strict theorem as a generic stability impossibility; admission here follows its actual threshold. MF-04 is a proposed finite-product property, whereas this entry asks for a decision procedure even if such a property fails.